differentiate into different cell types such as skin, muscle, bone, etc., (Figure 1.4).

Stem-cell therapy is the use of stem cells to treat or prevent a disease or condition. **Stem Cell Therapy** (SCT) is the treatment of various disorders, non-serious to life threatening, by using stem cells. These stem cells can be procured from a lot of different sources and used to potentially treat **more than 80 disorders**, including neuromuscular and degenerative disorders.

Hematopoietic disorders (Example: leukaemia, thallassemia, aplastic anemia, MDS, sickle cell anemia, storage disorders etc.) affect the bone marrow and manifest with various systemic complications. Stem cells from a donor (either from cord blood or bone marrow) are known to reconstitute the defective bone marrow and permanently overcome the disorder.

1.4 Molecular Biology and Genetic Engineering

Molecular biology- is the study of the structure, function & makeup of the molecular building blocks of life. It focuses on the interactions between the various system of a cell, including the interrelationship of DNA, RNA & Protein synthesis &how these interaction are regulated. Bioscience, Molecular biology closely interrelate with the fields of Biochemistry, Genetics & Cell biology.

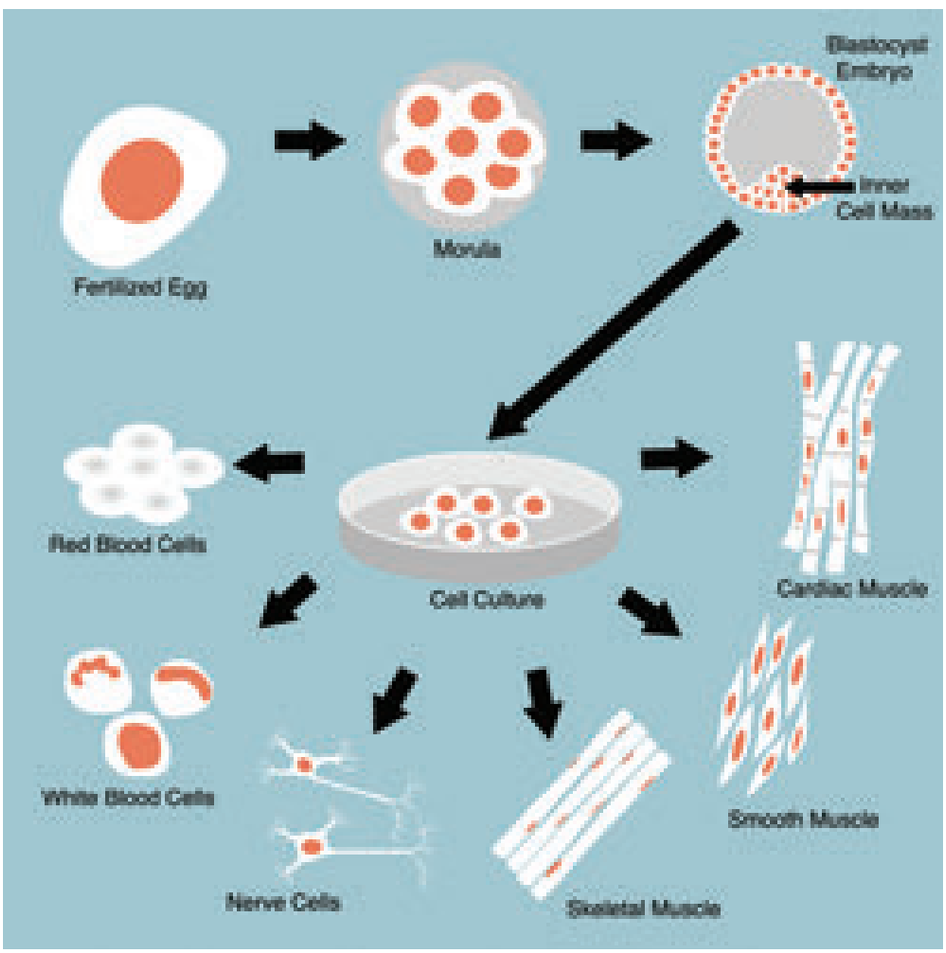


Figure 1.4: Stem cells

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